

$^{31}\text{P}(^6\text{Li},\text{d})$ 1979Es05

$^{31}\text{P} + \alpha \rightarrow ^{35}\text{Cl}$ from $J^\pi = 1/2^+$ ^{31}P ground state.

1979Es05,1978Es01: a 36-MeV $^6\text{Li}^{3+}$ beam at 150-400 nA was produced from the MP tandem accelerator at the University of Rochester and bombarded a $100\text{-}\mu\text{g}/\text{cm}^2$ cadmium phosphide Cd_3P_2 target. Outgoing deuterons were momentum analyzed by an Enge split-pole magnetic spectrograph and detected using Kodak NTB-50 nuclear emulsion plates with FWHM=50 keV. Measured $\sigma(E_{\text{d}},\theta)$. Deduced levels, J, π , L-transfers, spectroscopic factors from finite-range, LOLA-DWBA analysis of the measured $\sigma(\theta)$. Other: 1982Ch10 (finite-range DWBA calculations).

 ^{35}Cl Levels

Spectroscopic factor $g^*\text{S} = \sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}$, where $g = (2J_{\text{f}} + 1)/(2J_{\text{i}} + 1)$.

<u>E(level)[†]</u>	<u>Jπ[‡]</u>	<u>L[#]</u>	<u>S[#]</u>
0	3/2 ⁺	2	1.00
1220	1/2 ⁺	0	1.49
1760	5/2 ⁺	2	0.24
3160	7/2 ⁻	3	0.52

[†] From 1979Es05.

[‡] As given in 1979Es05 for extracting spectroscopic factors.

[#] From DWBA analysis of the measured $\sigma(\theta)$ in 1979Es05.